

Point elasticity

Example: Change in demand with respect to its own prices

$$x_1 = 10 - 2p_1$$

Price $p_1 = 1$, demand $x_1 = 8$

$$\frac{\text{Percentage change in demand}}{\text{Percentage change in price}} = \frac{p_1}{x_1} \frac{dx_1}{dp_1} = \frac{1}{8} (-2) = -0,25$$

Point Own-Price Elasticity

For example, suppose $p_i = a - bx_i$.
Then, $x_i = (a - p_i)/b$ and

$$\varepsilon_{x_1^*, p_1} = \frac{p_i}{(a - p_i)/b} \cdot \left(-\frac{1}{b}\right) = -\frac{p_i}{(a - p_i)}.$$

